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ORIGINAL ARTICLE

Stance detection in Arabic with a multi-dialectal cross-domain stance corpus

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Abstract

We present a cross-domain and multi-dialectal stance corpus for Arabic, covering the major dialect groups and four Arab regions. This research provides an important language resource for automating the task of stance detection in Dialectal Arabic while carefully considering the subtle differences in stance expression across various dialects. More than 4500 sentences in our corpus have been carefully annotated according to their stance with regard to a certain subject. We gathered sentences associated with two controversial topics for every region and we had at least two annotators annotate each sentence to indicate if the author is supporting, opposing, or neutral to the sentence's topic. Our corpus shows high balance between dialect and stance. About half of the sentences in each region are written in Modern Standard Arabic, while the other half are written in the specific dialect of that region. To evaluate our corpus, we performed a number of machine-learning experiments for the stance detection task. The best performance was achieved by AraBERT with an accuracy and an F1-score of 0.82. Furthermore, we trained and tested this model on the most similar state-of-the-art stance dataset, "MAWQIF". The comparison results demonstrate how crucial it is to maintain balance among the three stance classes in our dataset. In particular, the model scored better when using our stance corpus than when using the MAWQIF dataset especially for the "Neutral" stance class. Using our best performing model, we developed a Web-based demonstrator for stance detection in dialectal Arabic and we show its effectiveness in analyzing stance in the context of two real-world scenarios: product boycott in the Arab world and customer reviews of a soft drink company.

Keywords Natural language processing · Arabic · Machine learning · Corpus · Stance detection · Polarization · Arabic dialects

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1 Introduction

The word "stance" refers to the writer's expression of personal feelings, assessments, and judgments regarding a particular message or topic (Biber and Finegan 1988). Stance detection, a sub-task derived from sentiment analysis (Küçük and Can 2021), aims to ascertain the author's attitude towards a target, often categorizing it as against, in favor, or neutral (Li and Caragea 2019). This task serves as an important tool for analyzing and interpreting public opinions online (Cao et al. 2022).

With the widespread adoption of social media platforms globally, stance detection has gained increasing significance across various domains. In fact, it provides a key tool in analytical research to gauge public opinion on social, religious, and political matters, facilitates veracity checking applications, and assesses controversy levels on social media platforms (AlDayel and Magdy 2021).

Furthermore, stance detection is quite helpful in situations where individuals rely on online searches to make critical decisions, such as adopting a vegan lifestyle (Draws et al. 2023) or participating in a specific social movement. These choices could affect people's susceptibility to different points of view and possibly cause them to change their minds (Mckay et al. 2020; Flaxman et al. (2016)).

In addition, stance detection plays a crucial role in identifying polarization, which occurs when there is a strong inclination towards one party and a notable aversion or bias against the other (McCarty 2019). Thus, the detection of stance has been applied in various contexts, such as assessing polarization levels regarding topics like climate change on social media platforms (Upadhyaya et al. 2023). Consequently, the determination of stance serves as a foundation for evaluating the extent of polarization in a certain community or in a society.

At a broader level, polarization detection facilitates a deeper understanding of social media communities by pinpointing the presence of propaganda and radicalism (Bail et al. 2018). Once polarization is identified, it becomes feasible to initiate discussions on polarized topics or issues within the public sphere. This enables an exploration of the differing polarized viewpoints and, in certain scenarios, an attempt to bridge the split by establishing common ground.

In this work, we introduce MARASTA, which is a novel cross-domain and multi-dialectal stance corpus for Arabic, Our corpus contains over 4500 sentences annotated for their stance towards specific topics by at least two annotators. The corpus targets four significant dialectal regions in the Arab world: Maghreb, Egypt, the Gulf, and the Levant. In fact, Arabic dialects exhibit considerable variation, with each possessing distinct linguistic features, syntax, and vocabulary reflecting the diverse cultural and regional differences across the Arab world. For each region, we identified two controversial topics and then gathered sentences belonging to these topics. Then, annotators were asked to manually indicate whether the stance of each sentence was in favor of, against, or neutral towards the related topic. Half of the sentences for each topic are in the respective region's dialect, while the other half are in Modern Standard Arabic (MSA). Furthermore, our corpus maintains a balanced distribution across stance categories and dialects.

Our research methodology is represented graphically in Fig. 1. First, we started by developing our stance corpus. Then, we used our corpus and conducted several machine learning (ML) experiments in order to develop models for stance detection in Arabic. After that, we built a Web-based demonstrator for stance detection using the best performing ML model. Finally, we evaluated our stance detection model and demonstrator by applying them to two real-world scenarios: product boycott in the Arab world and customer reviews of a soft drink company.

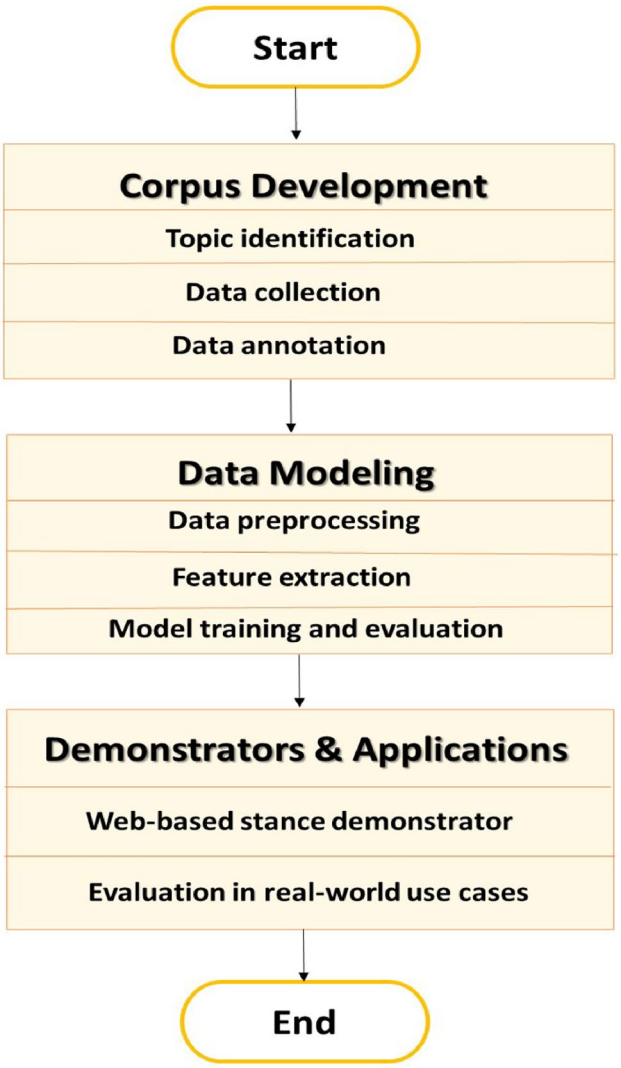

Fig. 1 Research methodology flow diagram

The remainder of this paper is organized as follows: Sect. 2 discusses related work. Section 3 introduces our proposed stance corpus and delineates the data collection and annotation processes. Section 4 presents the experiments we conducted to automatically detect stance using our corpus, along with an evaluation of our models' performance using a similar stance corpus. Section 5 presents a Web-based demonstrator that we built for stance detection in Arabic using our corpus and the best performing AraBERT model. Section 6 presents two real-world scenarios, in which our demonstrator is used to analyze stance in online content with regard to two trending topics. Section 7 discusses the strengths and limitations of our work as well as its practical and managerial implications. It also outlines directions for future research.

2 Related work

Stance detection, polarization, and sentiment analysis are interrelated fields, each targeting the identification of attitudes or opinions expressed by individuals, groups, or entities regarding specific topics or issues. While sentiment analysis primarily aims to detect the emotional tone of a text, stance detection dives deeper, seeking to determine the author's stance or position towards particular subjects. This section reviews current research and datasets related to these areas, with a special focus on works targeting the Arabic language.

Recent progress in sentiment analysis has significantly improved our ability to systematically analyze and interpret the emotional undertones present in a wide range of textual data. Studies in regards to this topic have addressed various domains, including healthcare sentiment analysis by Menaouer et al. (2022) and Mohammed et al. (2022); political sentiment analysis, explored by Dehghani and Yazdanparast (2023); and sentiment analysis related to fake news detection, investigated by Balshetwar and Rs (2023). Furthermore, researchers have used a variety of innovative methodologies to enhance sentiment classification accuracy in the Arabic language. For instance, Ombabi et al. (2020) used deep learning models such as CNN and LSTM networks to leverage Arabic's morphological richness, while Alayba et al. (2018) combined word embeddings with CNNs. Matrane et al. (2023) provided a comprehensive survey of sentiment analysis techniques and datasets, highlighting the necessity for more sophisticated models and large-scale annotated corpora.

Aspect-based sentiment analysis (ABSA) aims to identify sentiments related to specific aspects or features of entities, offering a more detailed understanding of opinions. This method enhances granularity by focusing on specific aspects within a text, rather than the overall sentiment. Recent studies, such as those by Liu et al. (2020) and Gu et al. (2023), have highlighted the potential of ABSA in providing more detailed insights into customer opinions and preferences. By analyzing a text into its component pieces and examining the sentiment expressed about each aspect, ABSA allows for a more refined understanding of user feedback, which is particularly valuable in domains like product reviews and social media analysis. Additionally, a study by Bansal and Kumar (2022) explored the application of ABSA in the healthcare sector, demonstrating its effectiveness in analyzing patient feedback to improve service quality and patient satisfaction. By comparing these approaches, we can better understand the distinctions and intersections between sentiment analysis and stance detection in Arabic text processing.

When it comes to stance detection and polarization, several works focused on this topic. For those works that developed ML models for stance detection in Arabic, we show in Table 1 their performance results.

The authors of Weber et al. (2013) created the first dataset aimed at detecting polarization in Arabic. This dataset includes Twitter users classified according to their polarization towards either Islamist or secular ideologies. The data collection process started with manually categorizing "seed users" as secularists or Islamists. By tracing retweet connections, the authors expanded the dataset to include 7,088 Twitter profiles, all located in Egypt, and automatically labeled them as either "Secularist" or "Islamist." The primary objective of this dataset was to investigate the polarized political environment in Egypt after 2011. However, this dataset is limited to a single topic and a single country.

The work of Haouari and Elsayed (2024) introduced the task of detecting the stance of authorities regarding rumors on Twitter, aiming to enhance existing rumor verification models. The authors developed the Authority Stance towards Rumors (AuSTR) dataset, which included 811 pairs of rumor and authority tweets in Arabic. Through extensive experiments using BERT-based models, the study found that a model trained solely on the AuSTR dataset achieved a macro-F1 score of 84% and an F1 score of 78% on debunking tweets. These results indicated that the stance of authorities—whether they supported, denied, or remained neutral

Table 1 Summary of related works in stance detection

Citation	Dataset used	Best model	Performance metric
Haouari and Elsayed (2024)	AuSTR	ARBERT	Macro F1-score: 0.84
Alqurashi (2022)	Their own dataset	SVM	Macro F1-score: 0.85, AUC: 0.95
Alturayeif et al. (2022)	MAWQIF	BERT-based models	Macro F1-score: 0.78
Alhindi et al. (2021)	AraStance	BERT-based models	Macro F1-score: 0.78, Accuracy: 0.85
Jaziriyan et al. (2021)	ExaASC	BERT	Macro F1-score: 0.70
Khouja (2020)	ANT	BERT-based models	F1-score: 0.76
Baly et al. (2018)	Their own dataset	ATHENE model	F1-score: 0.55, Accuracy: 0.80
Darwish et al. (2017)	The Islands & Islam Datasets	SVM	Macro-F1 score: 0.78; 0.79

towards a rumor—could significantly improve the effectiveness of rumor verification systems.

Alqurashi (2022) analyzed public opinions on distance education in Saudi Arabia during the 2020 academic year by examining 1,846,285 tweets. The researcher constructed his own dataset, covering the period from August 15, 2020—when the Ministry of Education announced the suspension of schools and the shift to distance learning—until the end of March 2021. After filtering out spam, advertisements, non-Arabic tweets, duplicates, and retweets, the cleaned dataset contained 464,410 tweets. A sample of 4,348 tweets, featuring 17,232 unique words, was used for learning. This study used classical and deep learning models, including RF, SVM, AdaBoost, MNB, CNN, and LSTM, across three stages: data collection, stance learning, and stance detection. Among these, the SVM classifier with TF–IDF as a feature extraction method achieved the highest performance, with F-measure and AUC scores of 85.9% and 95.1%, respectively.

The work by Darwish et al. (2017) presented a method for improving stance prediction in social media users by utilizing user interactions and non-topically relevant tweets to compute user similarities, rather than relying only on textual content. The authors leveraged the concept of homophily, where users with similar beliefs tend to connect, and employ a random graph walk technique to estimate user similarities based on interactions such as retweets, mentions, and hashtags. Their work was validated using two datasets: the Islam dataset, which included tweets related to the November 2016 ISIS attacks and consisted of 979 tweets tagged as expressing positive or negative views towards Muslims, and the Islands dataset, which featured Arabic tweets concerning the transfer of ownership of the islands of Tiran and Sanafir from Egypt to Saudi Arabia, comprising 48,445 tweets authored by 4,164 users. The proposed method achieved a macro-F1 score of 0.78 for the Islands dataset and 0.79 for the Islam dataset when using user similarity features.

The corpus developed by Baly et al. (2018) integrates fact-checking and stance detection by providing a set of annotated claim-article pairs, where each claim is linked to one or more articles. Initially, true and false claims were sourced from the Syrian fact-checking website "Verify" and the news website "Reuters". Using the Google Custom Search API, the authors then retrieved documents pertinent to these claims. For stance annotation, each claim-document pair was evaluated by 3–5 Arabic-speaking annotators. This corpus contains 3,042 claim-document pairs annotated with the document's stance towards the claim, categorized as "agree", "disagree", "discuss", or "unrelated". The authors applied stance detection models from the Fake News Challenge (FNC)[1] to their corpus, achieving a weighted accuracy of 55.6% and an $F_1 score$ of 41%. The corpus is entirely in MSA and does not include dialects. Additionally, it primarily focuses on the Syrian War and is notably unbalanced, with around 2000 pairs labeled as unrelated.

Khouja (2020) proposed an Arabic dataset that can be used jointly for claim verification and stance detection. To create the corpus, the author selected news titles from the Arabic News Texts (ANT) corpus,[2] which is a collection of Arabic news from multiple news media sources in the Middle East. Then, a crowd sourcing was used to produce both false and true claims from these titles. More specifically, annotators were asked to either paraphrase the original text or produce a contradictory claim. The resulting dataset, called *Ans*, consisted of 3786 pairs of original news titles and their modified versions. Each pair was annotated with one of three classes: agree, disagree, or other (in cases where there is insufficient information). To demonstrate the utility of the corpus for stance detection, the author evaluated two models: an RNN model that uses LSTM and the Pre-trained BERT model, achieving F1-scores of 37.8% and 76.7% respectively. This dataset has some limitations, including a restricted focus on MSA and an imbalanced class distribution, with a 63.4% sentence pairs categorized as "disagree", while only 34.4% of the pairs are labeled as "agree".

Alhindi et al. (2021) presented a multi-country and multi-domain dataset in Arabic called *AraStance*. Similar to the corpus created by Baly et al. (2018), this dataset consists of claim-article pairs that are labeled for both stance and veracity. False claims were collected from three fact-checking websites namely, ARAANEWS, DABEGAD,[3] and NORUMORS,[4] originating from the UAE, Egypt, and Saudi Arabia respectively. These claims were from 2012 to 2018 and covered various domains, including politics, sports, and health. On the other hand, true claims were collected from the Arabic Website of REUTERS.[5] Relevant documents were retrieved for each claim using various searches and the Google Search API. The final corpus comprised 4,063 claim-article pairs annotated as "agree", "disagree", "discuss", or "unrelated", stating the stance of the article towards the claim. The authors then conducted experiments using different pre-trained BERT models and the best performing model achieved an accuracy of 85% and a macro F1-score of 78%.

[1] http://fakenewschallenge.org.

[2] https://antcorpus.github.io/.

[3] http://dabegad.com/.

[4] http://norumors.net/.

[5] http://ara.reuters.com.

Jaziriyan et al. (2021) introduced the *ExaASC*, a target-based Arabic stance corpus focused on tweets and their corresponding replies. Data collection involved extracting argumentative tweets and their replies using Twitter's stream API. The target of the original tweet was identified by analyzing the content of the replies related to that tweet. The dataset includes 9566 tweet replies, each annotated by at least two native Arabic speakers. These annotations reflect the stance of each reply towards the target or an entity associated with the target, categorized as "favor", "against" or "none". The authors evaluated the corpus using different implementations of pre-trained BERT models and achieved a Macro $F_1 score$ of 70%. They found that models using only the target outperformed those using the entire source tweet sentence. However, a limitation of this corpus is its imbalance in terms of dialects due to the random extraction of tweets from different countries. Additionally, identifying the target in this dataset depends on the content of the source sentence.

Another work by Alturayeif et al. (2022) introduced a dataset that emphasizes stance detection using a target-based approach. This dataset includes 4,121 multi-dialectal Arabic sentences primarily collected from Twitter. It focuses on three targets: "COVID-19 vaccine", "digital transformation", and "women empowerment", each labeled as "favor", "against", or "none". Arabic annotators were hired through the Appen crowdsourcing platform to ensure cultural and linguistic accuracy. The dataset also includes three dimensions namely, "stance", "sentiment", and "sarcasm". Evaluation with four BERT-based models showed that the best model achieved a macro-$F_1 score$ of 78.89%. While the dataset is balanced in the total number of sentences for each target, it is imbalanced in the distribution of sentences per label.

To sum up, The majority of the datasets available are either unbalanced or focus on one topic only. Our goal is to develop a novel Arabic stance detection dataset that spans a diverse range of topics from different Arab nations, supports many Arabic dialects, and retains a well-balanced composition of content for each topic, class label, and dialect.

3 Overview of our stance corpus

In this section, we present our multi-dialectal Arabic Cross-domain Stance Corpus "MARASTA", outlining the steps involved in data collection and annotation. Unlike many existing corpora in this domain, our corpus addresses multiple topics and includes various Arabic dialects, ensuring a well-balanced representation.

Given the variety of Arabic dialects, we segmented the data collection task based on Arab regions with similar dialects. Specifically, we focused on the Maghreb region (with a particular emphasis on Tunisia), Egypt, the Levant Region (specifically Palestine and Jordan), and the Gulf region. Alongside these Arabic dialects, we also included MSA into the data collection process, which is the Arabic variant used as standard for official and generic communication. We collected over 1000 sentences for each of the four Arab regions, resulting in a total corpus size of 4657 meticulously balanced sentences.

Table 2 shows samples of sentences included in our corpus, accompanied by their corresponding English translations and stances towards their related topics.

3.1 Data collection

Our research team included six individuals from diverse Arab countries, each possessing a profound understanding of the controversial topics prevalent in their respective regions. For each region, we had at least two experts in that region. All team members were invited to suggest controversial topics for the regions they were familiar with. Subsequently, we verified for each suggested topic their relevance through searches on various social media platforms and Websites (such as Wikipedia and various online newspapers).

Our research team proposed three criteria for selecting the topics. Firstly, a topic had to be adequately controversial to ensure extensive discussions, with a substantial number of posts and comments reflecting diverse stances. Secondly, a topic could relate to current issues or those of relevance within the past 15 years. Thirdly, a topic could be related to a single country or multiple countries.

Following the initial compilation of controversial topic suggestions from all team members, we conducted an exploration of social media platforms to evaluate the availability of sentences related to these topics. Subsequently, for each Arab region, we identified the most highly rated topics based on the availability of data related to the topics and the three stances: in favor, against, neutral. If there were more than two topics with sufficient data available for one region, we had all team members vote to decide which two topics should be included in our corpus. We believe that involving more than one team member in the topic suggestion and having a vote by all team members on the topics to include reduced personal bias.

Once we decided on the topics to include, we started the process of collecting sentences related to these topics by using relevant seed keywords. Table 3 displays examples of the seed keywords we used to gather pertinent sentences for specific topics.

After choosing all topics, we started a manual collection of sentences that are related to the topic using the seed keywords mentioned earlier. A sentence must fulfill the following requirements so that we include it in our corpus: it

Table 2 Examples from each region of our corpus along with their translations and stances

Region	Topic	Example	Translation	Stance
Egypt	New Administrative Capital in Egypt	العاصمة الجديدة دي عاصمة الشيطان.	The new administrative capital is the capital of Satan	Against
	Egypt's 2013 Political Transition	الي صار في ٣٠ يونيو ٢٠١٣ كان ثورة لحماية الهوية المصرية.	What happened on June 30, 2013 was a revolution to protect Egyptian identity	Pro
Maghreb	Illegal Immigration to Europe	علاه ماناقفوش لبلادنا و كل بلاد تخدمها رجالها و بتشكر باب الحرقة.	Why don't we stand up for our country, when every country is served by its own men, and close the door of immigration	Against
	Tunisian General Labor Union	شنوا فيه البيان الجديد متاع الاتحاد؟	What is in the new statement of the labor union?	Neutral
Gulf Region	Fifa World Cup 2022	استقبلت الدوحة أكثر من مليون و اربع مئة الف زائر من أنحاء العالم خلال كأس العالم فيفا قطر ٢٠٢٢، التي انطلقت منافساتها في ٢٠ تشرين الثاني/ نوفمبر الماضي، و اختتمت يوم الأحد الماضي.	Doha has received more than 1.4 million visitors from around the world during the FIFA World Cup Qatar 2022, whose competitions began last November 20 and concluded last Sunday	Neutral
	Normalization With Israel	انا مع التطبيع مع عيال عمنا اليهود.	I support normalization with our Jewish cousins	Pro
Levant Region	Presence of Refugees in Jordan	السوريين اخوانا و الله ما حسينا بفرق بينانا و هيانا احنا و هني عايشين ع الحلوه والمره.	Syrians are our brothers, I swear we didn't feel the difference between us. We are both living together through the good and the bad	Pro
	Feminism and Women's Rights	المساواه و الحركة النسوية اهم سبب لتعاسة النسوان.	Equality and the feminist movement are the most important reason of women's misery	Against

must be in Arabic script (either in MSA or in the dialect of the designated region), relevant to the selected topic, and either express a stance (pro or against) or is neutral to that topic. Furthermore, a sentence must be grammatically correct. While we acknowledge that social media content often contains grammatical errors and informal language, our approach was to prioritize clarity and coherence over strict grammatical correctness. Sentences with minor grammatical errors or informal structures were still included if they were comprehensible and reflected genuine language use. We only excluded sentences with severe grammatical issues that impeded understanding or could potentially confuse our annotation process and model training.

Due to the time-consuming nature of manual data collection from various social media platforms and websites, we opted to also use social media platforms offering API access. We developed Python scripts for data collection from X (previously Twitter) and YouTube. The Python script for the X API requires the user to have a set of keywords related to the topic stored in a text file. The script then reads the text file and searches for tweets containing the specified keywords. Subsequently, the script generates a comma-separated value (.csv) file containing the relevant tweets.

For YouTube API, the script uses the conventional YouTube search method. Users initiate a search for videos related to a specific keyword, and the script retrieves links to relevant videos. Users are prompted to input the links of videos they deem relevant, and the comments from these videos are compiled and saved into a.csv file. Users are then required to review these files and transfer the pertinent sentences to the shared workbook containing our corpus data.

3.2 Corpus balancing

To ensure the balance of our stance corpus, we implemented two rules. Firstly, for every topic in each region, we aimed to have a minimum of 504 sentences, with 168 representing each stance class ("pro" for expressing favor, "neutral" for expressing no stance, and "against" for expressing opposition). Secondly, for each stance class, half of the sentences were required to be in MSA, while the remaining half should be in the respective regional dialect.

To enforce these rules, we developed dashboards for each region, which dynamically update as data is entered. These dashboards track the total number of sentences per topic, the distribution of sentences per stance class, and the proportion of sentences in each dialect. Users can filter the dashboards by topic, dialect, and stance class.

Table 4 displays the final count of sentences and their distribution per topic, dialect, and stance. This table reflects the outcomes after annotation and discussion meetings. As a result, some sentences were excluded, leading to a final

Table 3 Examples of seed keywords from each region's dialect and topic

Region	Topic	Keywords	Translation
Egypt	New Administrative Capital in Egypt	العاصمة الإدارية الجديدة; العاصمة الجديدة; العاصمة الادارية;	The New Administrative Capital; The New capital; The Administrative Capital
Maghreb	Illegal Immigration to Europe	الحرقة; الهجرة الغير نظامية; الهجرة السرية: أوروبا; إيطاليا	Immigration; Irregular Immigration; Clandestine Immigration: Europe; Italy
Gulf Region	Fifa World Cup 2022	كأس العالم في قطر; تنظيم كأس العالم قطر; فيفا ٢٠٢٢; فيفا قطر; مونديال قطر	World Cup in Qatar; The Organization of Qatar's World Cup; FIFA 2022; FIFA Qatar; Qatar's Mundial
Levant Region	Feminism and Women's Rights	النسويات؛ الحركة النسوية؛ الاحتجاجات النسوية؛ الذكورية	Feminists; The Feminist Movement; Feminist Protests; Masculinity

Table 4 Corpus statistics and sentence distribution

Region	Topic	Sentence count	Dialect distribution		Stance		
			MSA (%)	Dialect (%)	Pro	Neutral	Against
Egypt	New Administrative Capital in Egypt	494	50.6	49.4	156	157	181
	Egypt's 2013 Political Transition	480	56.7	43.3	165	146	169
Maghreb	Illegal Immigration to Europe	590	55.8	44.2	185	174	231
	Tunisian General Labor Union	713	65.2	34.8	213	201	299
Gulf Region	Fifa World Cup 2022	528	54.0	46.0	243	134	151
	Normalization With Israel	577	55.5	44.5	171	208	198
Levant Region	Presence of Refugees in Jordan	628	44.2	55.8	237	163	228
	Feminism and Women's Rights	647	46.1	53.9	203	183	261

Fig. 2 Word cloud visualization of MARASTA corpus

sentence count slightly lower than originally intended. The Word cloud in Fig. 2 visually represents the proposed corpus, with each word sized according to its frequency within the dataset, thereby highlighting the most frequent terms and concepts.

3.3 Corpus annotation

Our stance corpus was annotated by a team of proficient native Arabic speakers recruited as part-time staff from the four designated regions. Following the footsteps of Conforti et al. (2020), each sentence underwent blind annotation by two annotators. Both annotators were tasked with determining the stance of the sentence towards the given topic for each topic/sentence pair, with detailed instructions provided. Following these initial rounds of annotation, a script was used to identify conflicts where the first and second annotators provided different annotations. Conflicts were then resolved by involving a third annotator. If two out of the three annotations matched, they were considered as the final annotation. However, if all three annotations mismatched or if a sentence was deemed vague or unrelated to the topic by the second or third annotator, it was labeled as "discuss", prompting a discussion meeting attended by all three annotators involved. During the discussion meeting, conflicted sentences were deliberated upon, and a majority vote determined the final annotation. If all three annotators agreed that a certain sentence was unclear or irrelevant to the topic, it was eliminated from the corpus, resulting in the final numbers presented in Table 4. Throughout the annotation process, we ensured that annotators were proficient in the dialect of the sentences they were assigned.

To evaluate the annotation's quality, we calculated the inter-annotator agreement using Cohen's Kappa. Given the presence of two primary annotators, we used Cohen's Kappa coefficient, as illustrated in Table 5 for the various regions. All kappa scores exceeded 0.80.

To analyze the collected data from different social media platforms and Websites, we performed statistical evaluations

Table 5 Corpus' inter-annotator agreement metrics using Cohen's Kappa

Dialect	Cohen's Kappa
Egyptian	0.8157
Maghreb	0.8921
Gulf Region	0.8041
Levant Region	0.8265
Overall	0.8376

for each topic. We calculated the percentage distribution for each stance class (pro, against, neutral) across them. This approach helped us understand the stance tendencies specific to each source for the various topics and dialects. Table 6 summarizes these statistics for each dialect and its related topics. This table presents a detailed analysis of stance distributions across the various topics for the studied dialects on different platforms. It is noteworthy that when referring to each dialect, it includes both the region's dialect and MSA, ensuring a comprehensive representation of opinions and discussions within the respective Arabic-speaking communities.

As shown in Table 6, Twitter (or X) emerged as the most active platform for our data collection, showing significant discussions and a balanced distribution of stances across most topics. For instance, the Egyptian dialect's discussions on the New Administrative Capital and Egypt's 2013 Political Transition reveal active engagement on Twitter. In contrast, YouTube interactions, although fewer, exhibit strong 'pro' stances for many topics, such as the presence of refugees in Jordan for the Levantine dialect. Facebook demonstrates a high level of engagement, particularly in the Maghrebi dialect's discussions on the Tunisian General Labor Union and illegal immigration, showing strong 'pro' and 'against' stances. Websites, while having lower interaction levels, generally reflect a more 'neutral' stance. Instagram, included for the Gulf dialect topics, shows balanced

Table 6 Stance Distribution by topic and platform for the different dialects

Dialect	Topic	Platform	Neutral	Pro	Against
Egyptian	New Administrative Capital in Egypt	Twitter	131	114	160
		YouTube	14	38	21
		Websites	12	4	0
	Egypt's 2013 Political Transition	Twitter	32	94	80
		YouTube	23	71	88
		Websites	91	0	1
Maghrebi	Tunisian General Labor Union	Twitter	44	18	26
		YouTube	16	4	59
		Facebook	38	169	205
		Websites	104	22	9
	llegal Immigration to Europe	Twitter	55	16	46
		YouTube	27	33	73
		Facebook	22	122	71
		Websites	70	14	41
Levantine	Presence of Refugees in Jordan	Twitter	35	40	107
		YouTube	11	192	118
		Facebook	0	1	3
		Websites	117	4	0
	Feminism and Women's Rights	Twitter	93	112	54
		YouTube	11	71	62
		Facebook	9	16	144
		Websites	70	4	1
Gulf	Fifa World Cup 2022	Twitter	83	168	86
		YouTube	13	36	11
		Instagram	32	37	31
		Websites	6	2	23
	Normalization With Israel	Twitter	133	81	123
		YouTube	60	82	55
		Instagram	6	6	10
		Facebook	7	0	7
		Websites	2	2	3

stances but with less interaction. The analysis reveals that Twitter was the most engaging platform across all dialects and topics, serving as a primary medium for public discourse. On the other hand, websites typically exhibit a 'neutral' stance, suggesting they are more information-centric rather than platforms for public debate.

4 Experiments

This section presents the methodology and results of our machine learning (ML) experiments conducted to evaluate and test our corpus for the task of stance detection.

4.1 Experimental setup

Feature extraction with traditional ML approaches has been widely employed for the task of stance detection, including the use of Support Vector Machines, as proposed by Bar-Haim et al. (2017) and Mohammad et al. (2017), as well as Logistic Regression (Simaki et al. 2017), Random Forest (HaCohen-Kerner et al. 2017), and Decision Trees (Addawood et al. 2017). Artificial Neural Networks (ANNs) were also used in several studies, particularly classifiers based on Multi-Layer Perceptron (MLP), such as the work proposed by Zhang et al. (2018). These feature-based learning systems are typically used with predefined features such as character and word n-grams, POS tags, and hashtags to train the classifiers. They have been widely used in both earlier and recent work on stance detection. Besides, it is a typical practice in stance detection studies based on traditional feature-based machine learning algorithms to employ and test multiple approaches and compare them with each other (Küçük and Can 2020). More recent studies have increasingly adopted deep learning algorithms and large language models such as Google's Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al. 2018).

We first split our dataset into training and testing sets with a 75:25 ratio, comprising 3492 samples for training and 1165 samples for testing. Each sample extracted from the dataset comprises a topic name, a corresponding sentence, and a classification label indicating the stance of that sentence towards the topic. To pre-process the text data, we removed URLs, emails, stop words, punctuation, and non-Arabic characters. Furthermore, we conducted word stemming, which reduces words to their stem or base form. This process merges several word forms into a single representation, effectively reducing the dimensionality of the feature space. In fact, this pre-processing step led to improved performance results.

The proposed model is designed to determine the stance of a given sentence towards a specified topic, whether it is in favor, against, or neutral. Since ML models cannot understand text inputs directly, we used two feature extraction techniques: term frequency-inverse document frequency (TF–IDF) and bag-of-words (BoW). TF–IDF provides weighted features to increase performance. It calculates the weight of each feature in a document using the product of term frequency (TF) and inverse document frequency (IDF). BoW is mostly employed in text classification tasks where the occurrence of each word in a document is used to train the model. BoW builds a vocabulary of all unique words and their occurrence frequencies in all documents to train learning models. Besides, we propose a concatenation of BoW and TF–IDF features. In order to identify the most effective approach for our task, we built our classifier with a variety of ML methods namely, Support Vector Machines (SVM), Logistic Regression (LR), Random Forest (RF), Decision Trees (DT) and the MLP. These techniques are very different from each other, motivating an exploration of their respective accuracies in detecting stance within our Arabic corpus. They were conducted using Google Colab,[6] which served as the cloud-based computing infrastructure. It provided the essential computational resources required for training and evaluating the models. We used the *pandas* package to manipulate and analyze the data, the *nltk* package during the preprocessing phase to do tasks such as tokenization and stop word removal, and the *sklearn* package for training and evaluating the classifiers. Additionally, we used in our experiments the pre-trained language model AraBERT.

4.2 Results

To evaluate the performance of each model with our dataset, we used accuracy and $F_1 score$ as evaluation metrics. We also generated a classification report, which provides a detailed analysis of the model's performance for each class (pro, against, and neutral). This report helped us understand which models were better suited for our task. In addition, we analyzed the performance of each extracted feature to identify the most suitable feature for each model. Noting that during the pre-processing step, we removed stop words but we observed that this harmed the performance. This can be explained by the importance of some stop words, such as negation words, in determining the stance of a sentence within the context of our task. Based on that, we decided to skip the stop-word removal.

In the following, we present the stance detection results of the various ML models using BoW, TF–IDF, and the concatenation of TF–IDF and BoW features. Additionally, we present the results using the AraBERT language model.

[6] https://colab.research.google.com/.

4.2.1 Results with TF–IDF

TF–IDF calculates the weight of each feature in a document by multiplying term frequency (TF) and inverse document frequency (IDF).

Term Frequency (TF). The frequency of a feature in a document is denoted by TF, which is determined by the document's length. It can be defined as:

$$\mathrm{tf}(t,d) = \frac{\text{Frequency of term } t \text{ in document } d}{\text{Total number of terms in document } d}$$

This formula calculates the term frequency of a term t in a document d by dividing the frequency of term t in document d by the total number of terms in this document d. It provides a measure of how often a term appears in a document relative to the document's overall word count.

Inverse Document Frequency (IDF). IDF determines the amount to which a term is informative in a document for model training. It can be calculated as:

$$\mathrm{idf}(t,D) = \log\left(\frac{\text{Total number of documents in corpus } D}{\text{Number of documents containing term } t+1}\right)$$

The IDF formula calculates the inverse document frequency of a term t in a corpus D by taking the logarithm of the ratio between the total number of documents in the corpus and the number of documents containing the term t, with a smoothing term added to avoid division by zero. It measures the rarity of a term across the corpus, with less common terms having higher IDF values. For instance, stopwords have low IDF value.

Term Frequency-Inverse Document Frequency (TF–IDF). Finally, TF–IDF can be defined as:

$$\text{TF-IDF}(t,d,D) = \mathrm{tf}(t,d) \times \mathrm{idf}(t,D)$$

The TF–IDF formula combines the term frequency (TF) and inverse document frequency (IDF) to measure the importance of a term t in a document d relative to a corpus D. TF–IDF helps to identify terms that are both frequent within a document and rare across the corpus, thus emphasizing their significance in representing the content of the document.

Results. The TF–IDF results with the used classifiers are shown in Table 7. The SVM classifier achieved the best performance, with an accuracy and $F_1 score$ of 0.66. The confusion matrix displayed in Fig. 3 shows the performance of this model. In total, it correctly predicted the stances of 764 sentences and it made 401 incorrect predictions.

4.2.2 Results with BOW

The BoW is the most basic feature extraction technique used in natural language processing and information retrieval

Table 7 Performance comparison of machine learning models for stance detection on MARASTA Corpus using TF–IDF

Model	Accuracy	Class	Precision	Recall	$F_1 score$
SVM	**0.66**	Neutral	0.61	0.62	0.61
		Pro	0.75	0.60	0.67
		Against	0.64	0.76	0.70
		Macro avg	0.67	0.66	**0.66**
		Weighted avg	0.67	0.66	**0.66**
LR	0.65	Neutral	0.59	0.64	0.62
		Pro	0.70	0.60	0.65
		Against	0.66	0.70	0.68
		Macro avg	0.65	0.65	0.65
		Weighted avg	0.65	0.65	0.65
RF	0.64	Neutral	0.60	0.57	0.59
		Pro	0.68	0.63	0.65
		Against	0.65	0.71	0.68
		Macro avg	0.64	0.64	0.64
		Weighted avg	0.64	0.64	0.64
DT	0.54	Neutral	0.47	0.52	0.49
		Pro	0.56	0.50	0.53
		Against	0.57	0.59	0.58
		Macro avg	0.54	0.54	0.54
		Weighted avg	0.54	0.54	0.54
MLP	0.62	Neutral	0.56	0.65	0.60
		Pro	0.65	0.61	0.63
		Against	0.66	0.62	0.64
		Macro avg	0.62	0.63	0.62
		Weighted avg	0.63	0.62	0.62

The best accuracy and F1 scores obtained are highlighted in bold font

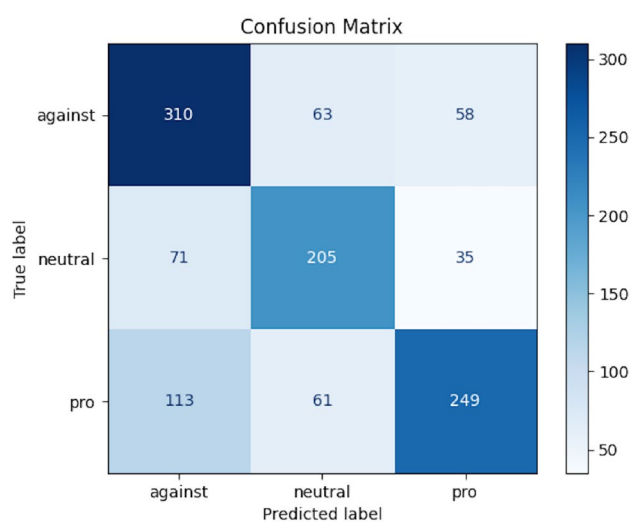

Fig. 3 Confusion matrix with SVM using TF–IDF

Table 8 Performance comparison of machine learning models for stance detection on MARASTA Corpus using BOW

Model	Accuracy	Class	Precision	Recall	$F_1 score$
SVM	0.62	Neutral	0.58	0.58	0.58
		Pro	0.71	0.50	0.59
		Against	0.59	0.76	0.67
		Macro avg	0.63	0.61	0.61
		Weighted avg	0.63	0.62	0.61
LR	**0.64**	Neutral	0.57	0.64	0.60
		Pro	0.68	0.61	0.64
		Against	0.65	0.66	0.66
		Macro avg	0.63	0.64	**0.63**
		Weighted avg	0.64	0.64	**0.64**
RF	0.63	Neutral	0.60	0.59	0.60
		Pro	0.67	0.64	0.65
		Against	0.63	0.66	0.64
		Macro avg	0.63	0.63	0.63
		Weighted avg	0.63	0.63	0.63
DT	0.55	Neutral	0.48	0.60	0.53
		Pro	0.59	0.52	0.55
		Against	0.58	0.55	0.56
		Macro avg	0.55	0.55	0.55
		Weighted avg	0.56	0.55	0.55
MLP	0.61	Neutral	0.56	0.61	0.58
		Pro	0.62	0.61	0.62
		Against	0.65	0.62	0.64
		Macro avg	0.61	0.61	0.61
		Weighted avg	0.62	0.61	0.62

The best accuracy and F1 scores obtained are highlighted in bold font

tasks. BoW is mostly used in text classification tasks where model training is based on the frequency of each word in a document. It creates a vocabulary consisting of all unique terms and the occurrence frequencies in all documents, which is used to train ML models.

The BoW results with the used classifiers are given in Table 8. The LR classifier achieved the best performance, with an accuracy of 0.64, a macro-average $F_1 score$ of 0.63, and a weighted average $F_1 score$ of 0.64. The confusion matrix given in Fig. 4 shows the performance of this model. In total, it correctly predicts the stances of 754 sentences, while making 411 incorrect predictions.

4.2.3 TF–IDF and BoW concatenation

We conducted further experiments by concatenating Bag-of-Words (BoW) and TF–IDF features.

The results of the different classifiers with the use of the concatenation of TF–IDF and BoW are given in Table 9. The LR classifier achieved the best performance, with an accuracy and $F_1 score$ of 0.64. The confusion matrix given

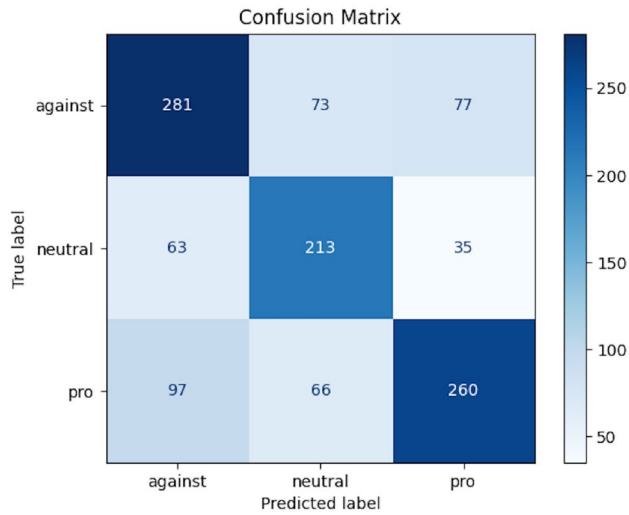

Fig. 4 Confusion matrix with LR using BOW

Table 9 Performance comparison of machine learning models for stance detection on MARASTA Corpus using TF–IDF and BoW concatenation

Model	Accuracy	Class	Precision	Recall	$F_1 score$
SVM	0.63	Neutral	0.58	0.59	0.59
		Pro	0.71	0.52	0.60
		Against	0.61	0.76	0.68
		Macro avg	0.63	0.62	0.62
		Weighted avg	0.64	0.63	0.62
LR	**0.64**	Neutral	0.58	0.64	0.61
		Pro	0.68	0.60	0.64
		Against	0.65	0.67	0.66
		Macro avg	0.64	0.64	**0.64**
		Weighted avg	0.64	0.64	**0.64**
RF	0.63	Neutral	0.59	0.60	0.60
		Pro	0.66	0.60	0.63
		Against	0.63	0.69	0.66
		Macro avg	0.63	0.63	0.63
		Weighted avg	0.63	0.63	0.63
DT	0.58	Neutral	0.53	0.60	0.56
		Pro	0.60	0.54	0.57
		Against	0.59	0.59	0.59
		Macro avg	0.57	0.58	0.57
		Weighted avg	0.58	0.58	0.58
MLP	0.62	Neutral	0.56	0.64	0.59
		Pro	0.63	0.61	0.62
		Against	0.66	0.62	0.64
		Macro avg	0.62	0.62	0.62
		Weighted avg	0.62	0.62	0.62

The best accuracy and F1 scores obtained are highlighted in bold font

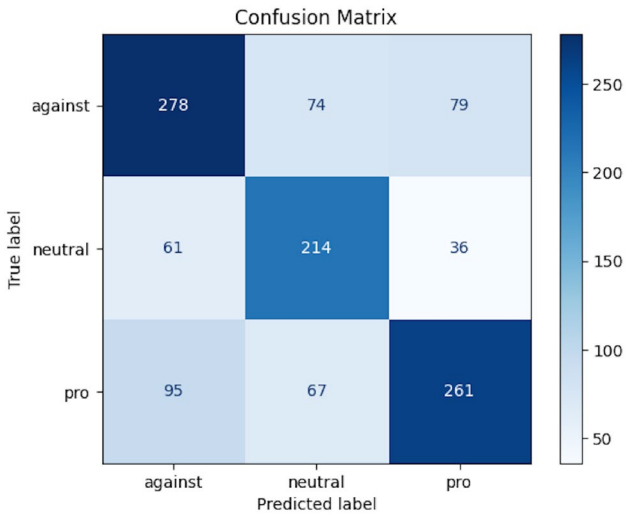

Fig. 5 Confusion matrix with LR using TF–IDF and BoW concatenation

Table 10 Initial performance results for stance detection using AraBERT

Epoch	Accuracy	Class	Precision	Recall	$F_1 score$
Epoch 1	0.76	Neutral	0.69	0.82	0.75
		Pro	0.88	0.67	0.76
		Against	0.74	0.81	0.77
		Macro avg	0.77	0.77	0.76
		Weighted avg	0.78	0.76	0.76
Epoch 2	0.79	Neutral	0.74	0.80	0.76
		Pro	0.85	0.77	0.81
		Against	0.78	0.80	0.79
		Macro avg	0.79	0.79	0.79
		Weighted avg	0.79	0.79	0.79
Average	0.77	Neutral	0.71	0.81	0.75
		Pro	0.86	0.72	0.78
		Against	0.76	0.80	0.78
		Macro avg	0.78	0.78	0.77
		Weighted avg	0.78	0.77	0.77

in Fig. 5 shows the performance of this model. In total, it correctly predicts the stances of 753 sentences, while making 412 incorrect predictions. Based on these results, we observed that using only TF–IDF feature extraction yielded the best outcomes, while combining it with BoW decreased the performance of the models.

4.2.4 Results using AraBERT

We also tested our corpus using the AraBERT model proposed by Antoun et al. (2020), which is an Arabic pre-trained language model based on Google's Bert architecture and specifically designed for Arabic text processing. Built on the Transformer architecture, AraBERT offers a thorough contextual understanding of the nuances and semantics of the Arabic language. Its flexibility allows for fine-tuning on specific tasks, making it highly effective for adapting its representations to our stance detection task.

Using AraBERT led to significantly improved performance compared to our previous experiments. Training the model for two epochs, we achieved an accuracy and $F_1 score$ of 0.76 in the first epoch and further improvement to 0.79 in the second epoch. Table 10 shows performance metrics we obtained for each epoch along with their averages. Then, we increased the number of epochs to 10 epochs, leading to significant improvements in accuracy and F1-score metrics. This enhancement can be attributed to several factors. First, extending the number of epochs provides the model with more opportunities to learn from the data, allowing for a deeper understanding of the underlying patterns and relationships. With additional epochs, the model can better adjust its weights and biases, resulting in reduced training loss and improved accuracy and F1-score. More training epochs also enable the model to capture complex features and subtle distinctions in the data. To mitigate the risk of overfitting, we employed early stopping based on validation loss. This technique monitors the model's performance on a validation set and halts training when improvements decline. So training was stopped when an an overfitting was observed. This approach ensured that we achieved optimal model performance while preventing unnecessary overfitting (see Figs. 6 and 7). The best accuracy and F1-score we achieved were 0.82 (see Table 11).

Next, we aimed to evaluate the model's performance across the different Arabic dialects to determine its effectiveness for each specific dialect. We trained and tested the model separately for each dialect to determine its effectiveness in detecting stance within each linguistic variety. The obtained model's results are summarized in the Table 12. These results shows that for the Levantine dialect the model had the best overall performance, with an accuracy of 0.89 and strong $F_1 scores$ across all stance classes. For the Gulf dialect, the model performs the least effectively, particularly struggling with the "neutral" stance detection. For the Egyptian dialect, the model also struggles the most with the "pro" class, while for the Maghrebi dialect the model performs consistently well, especially in the "against" class. These results highlight the importance of considering dialect-specific nuances in stance detection tasks and demonstrate that the model performance can vary significantly across different dialects.

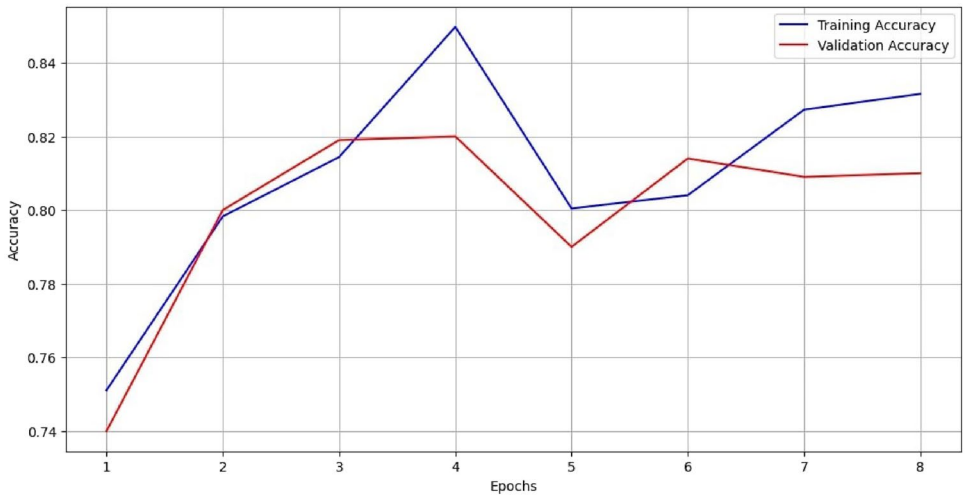

Fig. 6 Training and Validation accuracy graph of Our AraBERT Model for Stance Detection

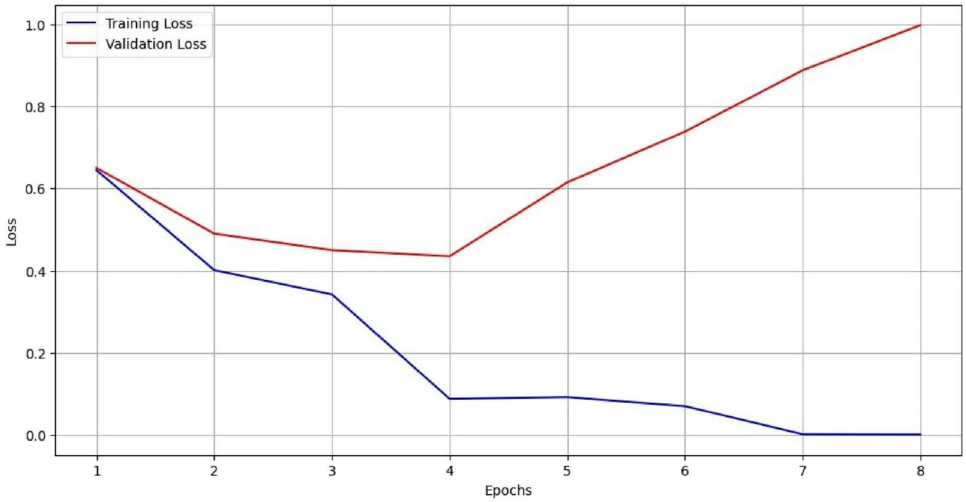

Fig. 7 Training and Validation loss graph of Our AraBERT Model for Stance Detection

Table 11 Performance results for stance detection using AraBERT

Accuracy	Class	Precision	Recall	$F_1 score$
0.82	Neutral	0.78	0.82	0.80
	Pro	0.86	0.81	0.83
	Against	0.81	0.84	0.82
	Macro avg	0.82	0.82	0.82
	Weighted avg	0.82	0.82	0.82

Table 12 AraBERT performance across different Arabic dialects

Dialect	Accuracy	Class	Precision	Recall	$F_1 score$
Egyptian	0.78	Neutral	0.86	0.79	0.82
		Pro	0.77	0.66	0.71
		Against	0.72	0.88	0.79
Maghrebi	0.84	Neutral	0.83	0.86	0.85
		Pro	0.76	0.88	0.81
		Against	0.93	0.80	0.86
Levantine	0.89	Neutral	0.95	0.85	0.90
		Pro	0.90	0.90	0.90
		Against	0.86	0.92	0.89
Gulf	0.75	Neutral	0.61	0.77	0.68
		Pro	0.82	0.72	0.77
		Against	0.84	0.77	0.80

4.3 Discussion

We conducted various experiments on our proposed corpus, exploring different feature extraction methods such as TF–IDF and BoW, as well as their combination. However, the concatenation of TF–IDF and BoW did not lead to improved results. Instead, using only TF–IDF provided better performance. Among the tested classifiers, the SVM performed the best with TF–IDF, yielding an accuracy

Table 13 Comparison of evaluation results between MARASTA and MAWQIF datasets for each stance class using AraBERT

Class	Metric	Dataset	
		MARASTA	MAWQIF
Neutral	Precision	0.78	0.56
	Recall	0.82	0.32
	$F_1 score$	0.80	0.41
Pro	Precision	0.86	0.87
	Recall	0.81	0.86
	$F_1 score$	0.83	0.86
Against	Precision	0.81	0.70
	Recall	0.84	0.81
	$F_1 score$	0.82	0.75
	Accuracy	0.82	0.79
	Overall $F_1 score$	0.82	0.69

and $F_1 score$ of 0.66. Further performance improvement was observed when using AraBERT, a model based on the Transformer architecture designed specifically for the Arabic language. AraBERT was fine-tuned on our stance detection task and trained for two epochs. This resulted in a notable increase in performance, with an accuracy and $F_1 score$ ranging from 0.76 in the first epoch to 0.79 in the second epoch. Notably, increasing the number of epochs further enhanced the performance of AraBERT, resulting in an accuracy and $F_1 score$ of 0.82. These results highlight the superiority of AraBERT in understanding and processing the nuances of Arabic, demonstrating its effectiveness and adaptability for our specific task.

To assess the effectiveness of our proposed methodology, we compared our model's performance metrics with the those reported in the literature (summarized in Table 1). Our model shows competitive performance, acknowledging the differences in datasets. For instance, Haouari and Elsayed (2024) achieved a macro F1-score of 0.84 using ARABERT on the AuSTR dataset, and Alqurashi (2022) reported a macro F1-score of 0.85 with SVM on their own dataset. Additionally, Alhindi et al. (2021) achieved a macro F1-score of 0.78 and an accuracy of 0.85 using using BERT-based model on the AraStance dataset, while Alturayeif et al. (2022) reported a macro F1-score of 0.78 on the MAWQIF dataset. Although our results are slightly lower than those reported by Alqurashi (2022) and Haouari and Elsayed (2024), they still reflect the strong potential of AraBERT on our dataset.

Among the state of the art corpora for Arabic stance detection, we found that the publically available MAWQIF dataset (Alturayeif et al. 2022) is the most comparable to our corpus as it considers the stance of sentences towards specific targets. Therefore, we used the MAWQIF dataset for comparison purposes and we tested it using the AraBERT model which demonstrated the best performance with our dataset. Table 13 presents a comprehensive comparison of evaluation metrics between our dataset "MARASTA" and "MAWQIF" for each stance class. Three key metrics, namely Precision, Recall, and $F_1 score$, are provided for each class. In addition, we included the overall accuracy and $F_1 score$ to provide a global perspective of model performance on both datasets.

Table 14 Comparison of class balance between MARASTA and MAWQIF

Class	MARASTA		MAWQIF	
	Train	Test	Train	Test
Neutral	1055	311	333	59
Pro	1150	423	2148	380
Against	1287	431	1021	180
All	3492	1165	3502	619
Total	4657		4121	

Examining the overall accuracy and $F_1 score$, we observe that AraBERT provides better results with our dataset than with the MAWQIF dataset with an accuracy of 0.82 compared to 0.79, and $F_1 score$ of 0.82 compared to 0.69. To better understand the performance results, it is important to delve into the metrics for individual stance classes. When used on our dataset "MARASTA", the AraBERT model demonstrates consistent performance across classes, with an $F_1 score$ of 0.80 for "neutral", 0.83 for "pro" and 0.82 for the "against" class. In contrast, when used with "MAWQIF" dataset, the model shows significant variation: an $F_1 score$ of 0.41 for the "neutral" class, 0.86 for the "pro" class, and 0.75 for the "against" class. These differences highlight the strengths of our dataset across all stance categories, showcasing consistent performance. While MAWQIF excels only in the "pro" and "against" classes, our dataset's balanced distribution of sentences among classes ensures the model is exposed to a diverse range of examples, allowing for better learning and generalization. Conversely, MAWQIF's unbalanced distribution may hinder the model's performance across the three stance categories.

Table 14 provides a comparison of the number of instances for each class in both the training and testing sets of the two datasets. It also offers insights into the distribution of instances across different classes in both datasets, allowing for a detailed examination of class balance.

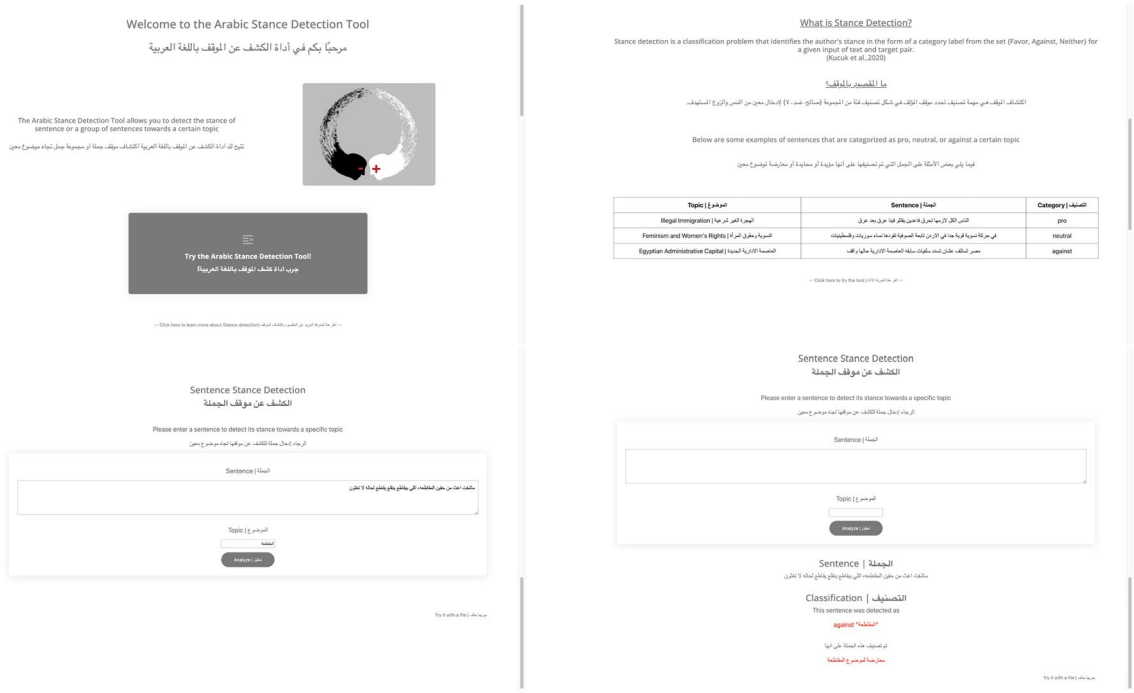

Fig. 8 Flow for stance detection in the tool

5 Stance detection web demonstrator

In this section, we discuss the architecture of our working Web-based demonstrator for stance detection along with its features and various sections.

5.1 Demonstrator architecture

Our stance detection demonstrator is a Web application that we developed using Python's Django framework. The Django framework is responsible for connecting the HTML, CSS, and JavaScript code, which together constitute the frontend of the demonstrator to the model, acting as a middleware in between. The model is saved within Django as a ".pkl" file using Python's Pickle module and is loaded as the server starts.

The data sent to the stance detection model consists of strictly Arabic sentences, i.e. we check the input and only Arabic sentences are allowed on input along with their topics. Alternatively, the demonstrator accepts a.csv file containing multiple sentences along with their respective topics. The data is firstly run through a wrangler that cleans it, then a vectorizer, which is dumped from the training data. After that, the vectorized data is sent to the fine-tuned AraBERT model on our proposed dataset, presented in the previous section, which determines the stance predictions and sends them to the frontend so that they are finally shown to the user.

In the case the input data is provided in file format, it runs through the same steps as above. Furthermore, a new file with an additional "Stance" column for the stance predictions is generated and dissected in the backend. The dissected file string is then sent to the frontend along with visual analytics about the newly generated column. JavaScript is used to re-generate the string into a.csv file that is automatically downloaded. A donut chart is also generated, showing statistics about the stance distributions, enabling stance and polarization analysis in large datasets.

This Web-based demonstrator is deployed on a Unix virtual machine, which is accessible at this URL.[7]

5.2 Demonstrator features and sections

Our Web-based stance demonstrator features two Web pages made up of four sections as shown in Fig. 8:

- The first section introduces the tool and its functionality.
- The second section defines stance detection as a task.
- The third section is for detecting the stance of a single sentence.
- The fourth section is for detecting the stances of multiple sentences (.csv file input and output).

[7] https://adhar.qatar.cmu.edu/polarization/.

Table 15 Examples of the sentences collected for the Product/Brand Boycott topic

Region's dialect	Tweet	Translation	Stance
Maghreb	ماعمري ما شفت أمرج من الناس المقاطعة. إذا تحب تقاطع قاطع من غير ما تمرجنا.	I have never seen more annoying people than those who boycott. If you boycott, do it alone without annoying us	Against
Gulf Region	مدري بس برأيي المقاطعة اختيارية	I don't know, but in my opinion, boycott is optional	Neutral
Egyptian	لا المقاطعه بقت اسلوب حياه خلاص الناس كلها بقت تكره تجيب اي حاجه داعمه للاحتلال	No, the boycott has become a way of life. That's it, everyone now hates to do anything that supports the occupation	Pro

Table 16 Examples of the sentences collected for "Kenza" customer reviews

Region's dialect	Tweet	Translation	Stance
Egyptian	كينزا دا معفن اوي و معرفش كيف الناس بتشربه	Kenza is so disgusting, I don't even know how people drink it	Against
Gulf Region	على كثر المدح اللي جاء على مشروب كينزا السعودي جاني فضول اجربه	With the amount praise that the Saudi Kenza drink received, I got curious to try it	Neutral
Levant Region	بصراحة كينزا بجنن وبديل ممتاز للمشروبات التانية	Honestly, Kenza is amazing and an excellent alternative to other drinks	Pro

Fig. 9 Input of stance detection demonstrator

The first section is the initial section that the user lands on when visiting this demonstrator. It explains the demonstrator's content and contains hyperlinks to the tool and the second section, which explains further the stance detection task. This section shows some examples from our dataset. The user may also use those examples to test the tool in the third section of the tool. To that end, the user is prompted to enter an Arabic sentence and a topic, and our model will detect and return the stance of that sentence towards the provided topic. In this section, there is also an hyperlink to the file input stance detection section, where the user is prompted to enter a file, and a new file will be generating with the "Stance" column. In this case, a donut chart is displayed with statistics about the stance distributions for the input file (i.e., how many sentences are in favor, how many sentences are against and how many sentences are neutral to the topic).

6 Evaluation in real-world use-case scenarios

A key objective of our stance detection demonstrator is to showcase the effectiveness of the fine-tuned AraBERT model on our dataset in detecting stances of sentences and topics even for those that were not included in our dataset.

To that end, we further evaluated our demonstrator using two controversial topics in the Arab World, which were not included in our corpus: "Boycott of Products/Brands supporting Israel during Gaza war in the Arab World" and "Customer reviews of a Gulf-based soft drink company called "Kenza"".

For each of the two topics, we collected a set of over 100 random sentences from online sources (examples shown in Tables 15 and 16). These sentences were

Fig. 10 Output of stance detection demonstrator

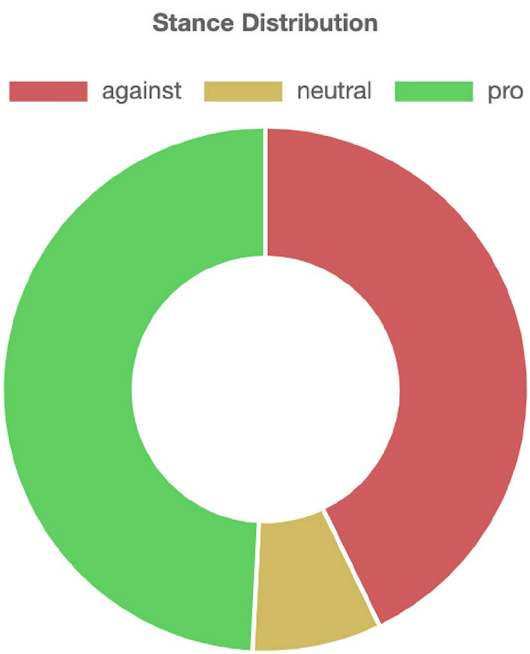

Fig. 11 Stance distribution for the products boycott topic

chosen to represent four Arabic dialects: Gulf, Levantine, Egyptian, and Maghrebi dialects, with a variety of stances within each group. These sentences were subsequently used to test our demonstrator's effectiveness and prediction accuracy.

To use our demonstrator, the user simply enters the sentence to analyze along with the corresponding topic as shown in Fig. 9. Once the user presses the button *analyze*, the demonstrator will use the fine-tuned AraBERT model with our MARASTA corpus to determine the stance of the sentence. Once a prediction is generated, the result will be displayed on the Web frontend with a green font for "in favor of", red for "against", and grey for "neutral" as shown for example in Fig. 10.

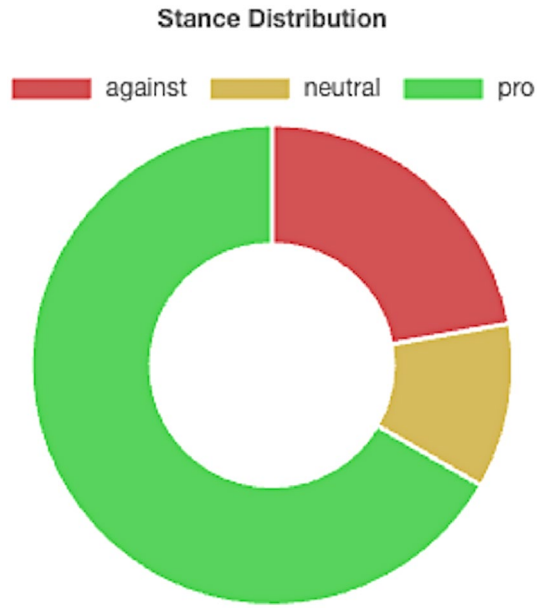

Fig. 12 Stance distribution results of Kenza's customer reviews

Additionally, the user can upload a file with multiple sentences organized in two columns: "sentence" and "topic". As described earlier, the demonstrator will then create a.csv file with an extra "stance" column that includes the predicted stance for each sentence. In this case, the demonstrator generates an interactive donut chart showing the stance distribution and the polarization levels for each topic as shown in Figs. 11 and 12.

6.1 Use-case scenario 1: boycott of products and brands

A product boycott is a consumer-driven effort to abstain from purchasing specific goods or services to express disapproval of certain actions or policies by a company, country, or government. One prominent boycott movement in Arab countries is the boycott of Israeli products, driven by different concerns related to the Israel-Palestine conflict and the current Gaza war. This use case scenario focuses on analyzing polarization in the Arab world around the topic of boycotting Israeli products and brands. In this scenario, our demonstrator allows governments and political leaders to better understand public opinion in their countries and regions.

After gathering 100 sentences on the boycott topic from various sources, we ensured they reflected diverse stances and dialects. Each sentence was manually annotated to accurately capture the related stance. We then uploaded a CSV file containing these sentences to our online demonstrator for analysis.

Our stance demonstrator processed the data and returned a new CSV file that included a column with the newly generated stances for each sentence. Additionally, it provided a donut chart illustrating the stance distribution for the boycott topic as shown in Fig. 11. This analysis revealed a higher polarization towards the "pro" stance compared to "against" and "neutral" stances. Specifically, the results showed 50% for the "pro" stance, 10% for the "neutral" stance, and 40% for the "against" stance. These findings align accurately with our manually annotated results, indicating the robustness of the analysis and confirming the good performance of our model even when used with external data from other domains.

6.2 Use-case scenario 2: customer reviews for the soft drink company "Kenza"

To cover a wider range of scenarios, we evaluated our stance detection demonstrator in a business context by analyzing customer reviews of products of a Gulf based soft drink company called "Kenza". We gathered approximately 100 sentences from X (formerly Twitter), ensuring they represented a variety of stances and dialects.

After manually annotating the data and using our tool to determine their stances, the results were as follows: 68% for the "pro" stance, 12% for the "neutral" stance, and 20% for the "against" stance as shown by the donut chart in Fig. 12. These findings indicate a significant majority of positive stances towards the company and its products. This high percentage of favorable stances suggests strong consumer satisfaction, which is a crucial key performance indicator for the company Kenza.

Furthermore, the presence of neutral and negative stances, though smaller in proportion, provides valuable insights into areas where the products of that company may need improvement or where customer expectations might not be fully met. This comprehensive analysis not only highlights the overall positive perception of "Kenza" by customers, but it also offers a detailed understanding of the public stance, enabling the company to make informed decisions and strategies.

Overall, the results give a positive indication for the company and would eliminate the need to conduct expensive surveys to seek customer feedback. The high approval rate can be leveraged in marketing and product development strategies to increase customer satisfaction and loyalty.

7 Conclusion

In this work, we presented MARASTA, a cross-domain and multi-dialectal Arabic stance corpus, covering four major regions in the Arab World: the Maghreb, Egypt, the Levant region, and the Gulf. Our corpus comprises over 4500 carefully balanced sentences across regions, dialects, and stances, focusing on two controversial topics for each region. Our experiments with various machine learning techniques, including the state-of-the-art AraBERT model, demonstrated the effectiveness of our corpus for stance detection tasks. The comparison with the MAWQIF dataset further highlighted the importance of maintaining a balanced distribution across stance classes.

The strengths of our work can be summarized as follows. First, our corpus is multi-dialectal. Unlike many existing corpora, MARASTA addresses multiple Arabic dialects, providing a more comprehensive representation of the linguistic diversity in the Arab world. Second, our corpus is balanced. In fact, it is well-balanced in terms of regions, dialects, and stances, which is crucial for developing robust and unbiased stance detection models. Third, we followed a rigorous annotation process for our corpus: Each sentence underwent at least two rounds of annotation, with a third round added in case of conflicts. Fourth, we developed a publicly available Web-based demonstrator allowing the community to access our stance detection functionality. We also applied our stance detection models to real-world scenarios, which involve different data compared to that used in training our models. The results obtained confirm strong cross-domain applicability of our approach.

The limitations of our work can be summarized as follows. First, our corpus size is rather small. While substantial, our corpus of 4500 sentences is relatively small compared to some large-scale datasets in other languages. Second, with approximately 500 sentences per topic, the word coverage per topic for each dialect might be limited. Third, some Arabic dialects are still not represented appropriately in our work such as Iraqi and Yemeni. Along the same lines, some regions, like the Maghreb, are represented by only one dialect (Tunisian in this case), which may not fully capture the linguistic diversity within that region.

This work has several important practical and managerial implications. Our stance detection tool enables organizations such as public entities or companies to efficiently analyze large volumes of Arabic text data to understand public opinion and sentiment on key issues, informing strategic decision-making. By quickly identifying polarizing topics and sentiment shifts, leaders and managers can proactively address potential crises or controversies before they escalate. As demonstrated in our real-world use cases, the tool provides valuable insights into public opinion on current topics such as product boycott. It also allows business managers to better understand customer attitudes towards their products and brands across Arabic-speaking markets. Understanding stance distributions, as represented graphically by our tools using donut charts, allows a better understanding of

stance distributions and polarization, which in turn enables more tailored and effective communication strategies in public communication, marketing, public relations, and stakeholder engagement. The multi-dialectal nature of our corpus enables pan-Arab analysis, crucial for organizations operating across multiple Arabic-speaking countries. Our Web-based demonstrator allows for continuous monitoring of public stance on emerging issues, enabling agile responses to changing sentiments. Furthermore, our corpus and tool provide valuable resources for academic and market researchers studying public opinion trends in the Arab world. By leveraging these implications, governments, companies, and organizations can make more informed decisions, improve stakeholder relations, and better navigate the complex landscape of public opinion in Arabic-speaking contexts.

Our future work should focus on expanding the corpus size, increasing the number of sentences per topic, and including more Arabic dialects. Additionally, further evaluation of our stance detection demonstrator in diverse real-world scenarios and the development of stance detection APIs for third-party integration would enhance the practical impact of this research.

Despite its limitations, MARASTA represents a significant contribution to Arabic natural language processing resources, particularly in the domain of stance detection. It provides a solid foundation for developing more sophisticated and culturally nuanced stance detection models, which can have wide-ranging applications in social media analysis, public opinion research, and targeted communication strategies across the Arabic-speaking world.

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Data availability Data will be shared soon with the academic research community.

Declarations

Competing interests The authors declare no competing interests.

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